While both products compete in almost all functionality, Solr is a favorite for large, static datasets working in big data ecosystems. Obviously, we have to run through prototypes and analysis to pick a product; the general trend is for projects that are integrating with a search engine for the first time to consider Elasticsearch due to its top-class documentation, community, and nearly no-hurdle startup. You must make a detailed comparison of your intended use cases for the search engine before adopting and embracing one.

Amazon OpenSearch

Elastic changed its licensing policy in 2021. The licensing, which applies to Elasticsearch release versions 7.11 and above, has been moved from open source to a dual license under an Elastic License and SSPL. This license allows the community to use the product for free, as expected, but managed service providers can no longer provide the products as services. There was a spat between Elastic and Amazon Web Services (AWS) when AWS created a forked version of Elasticsearch—called Open Distro for Elasticsearch—and offered it as a managed service. This spat led to the change in the license, which eventually led to OpenSearch's birth.

As Elastic moved from the open source licensing model to the SSPL model, a new product called OpenSearch (https://opensearch.org) was developed to fill the gaping hole left by the new licensing agreement. The base code for OpenSearch was created from the open source Elasticsearch and Kibana version 7.10.2. The product's first General Availability version 1.0 was released in July 2021. Watch out for OpenSearch becoming a competitor to Elasticsearch in the search engine space.

Now that we have a fair understanding of what a modern search engine is and the shape of the search landscape, let's jump into an overview of Elasticsearch.

1.4 Elasticsearch overview

Elasticsearch is an open source search and analytics engine. Developed in Java, it is an ultra-fast, highly available search engine built on the popular full-text library Apache Lucene (https://lucene.apache.org). Elasticsearch wraps around the powerful functionality of Lucene by providing a distributed system with RESTful interfaces. Lucene is the powerhouse of Elasticsearch, and Kibana is the administrative UI to manage and work with Elasticsearch. We work with Kibana's code editor (Dev Tools) throughout this book.

Full-text searching is where Elasticsearch excels as a modern search engine. It can retrieve relevant documents in response to a user's search criteria at an awesome speed. We can search for exact terms, too, like keywords, dates, or a range of numbers or dates. Elasticsearch is packed with top-notch features such as relevancy, "did-you-mean" suggestions, auto-completion, fuzzy and geospatial searching, highlighting, and more.

In addition to being a frontrunner in providing near-real-time search capabilities, Elasticsearch stands tall in statistical aggregations on big data. Of course, we must consider the use case before embracing the product, as Elasticsearch may not be the best fit for every use case (refer to section 1.4.3 to learn about the use cases). Out of the box, Elasticsearch also boasts commendable features such as application performance